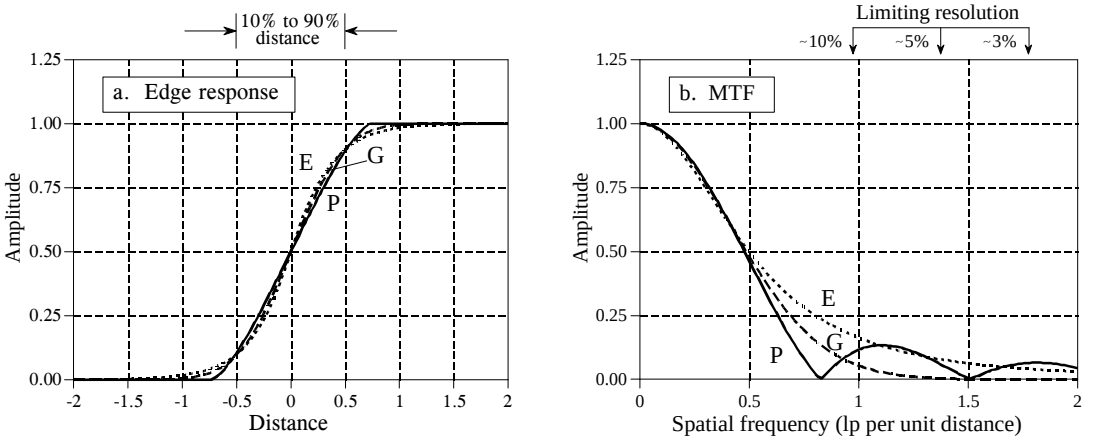

FIGURE 25-4
Edge response and MTF. Figure (a) shows the edge responses of three PSFs: (P) pillbox, (G) Gaussian, and (E) exponential. Each edge response has a 10% to 90% rise distance of 1 unit. Figure (b) shows the corresponding MTF curves, which are similar above the 10% level. *Limiting resolution* is a vague term indicating the frequency where the MTF has an amplitude of 3% to 10%.

The fifth advantage is that similar edge responses have similar MTF curves, as shown in Figs. 25-4 (a) and (b). This allows us to easily convert between the two measurements. In particular, a system that has a 10%-90% edge response of *x* distance, has a limiting resolution (10% contrast) of about 1 line pair per *x* distance. The units of the "distance" will depend on the type of system being dealt with. For example, consider three different imaging systems that have 10%-90% edge responses of 0.05 mm, 0.2 milliradian and 3.3 pixels. The 10% contrast level on the corresponding MTF curves will occur at about: 20 lp/mm, 5 lp/milliradian and 0.33 lp/pixel, respectively.

Figure 25-5 illustrates the mathematical relationship between the PSF and the LSF. Figure (a) shows a pillbox PSF, a circular area of value 1, displayed as white, surrounded by a region of all zeros, displayed as gray. A profile of the PSF (i.e., the pixel values along a line drawn across the center of the image) will be a rectangular pulse. Figure (b) shows the corresponding LSF. As shown, the LSF is mathematically equal to the *integrated profile* of the PSF. This is found by sweeping across the image in some direction, as illustrated by the rays (arrows). Each value in the integrated profile is the *sum* of the pixel values along the corresponding ray.

In this example where the rays are vertical, each point in the integrated profile is found by adding all the pixel values in each column. This corresponds to the LSF of a line that is *vertical* in the image. The LSF of a line that is *horizontal* in the image is found by summing all of the pixel values in each *row*. For continuous images these concepts are the same, but the summations are replaced by integrals.

As shown in this example, the LSF can be directly calculated from the PSF. However, the PSF cannot always be calculated from the LSF. This is because the PSF contains information about the spatial resolution in *all directions*, while the LSF is limited to only one specific direction. A system